

# Obed Junias

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## RESEARCH STATEMENT

I am a graduate researcher exploring **Commonsense Reasoning, Neuro-symbolic NLP, and Responsible ML**. My research focuses on developing reasoning frameworks that integrate structured logic with neural representations, with the goal of improving the interpretability, factual consistency, and fairness of LLMs. I aspire to contribute to foundational AI research at the intersection of cognition, reasoning, and ethical deployment.

## EDUCATION

**M.S. in Computer Science, University of Colorado Boulder** Aug 2024 – May 2026  
*Courses:* NLP, Deep Natural Language Understanding, Neurosymbolic NLP Methods (GPA: 4.0/4.0)

**B.E. in Computer Science, BMS College of Engineering** Aug 2017 – Aug 2021  
*Relevant Courses:* Algorithms, Databases, Operating Systems, Machine Learning

## CURRENT RESEARCH PROJECTS AND EXPERIENCE

**Commonsense Reasoning with Logical Entailment Trees, *BLAST Lab*** June 2025 – Present  
*Advisor:* [Dr. Maria L. Pacheco](#), CU Boulder

- Investigating evaluation of multi-fact commonsense QA using structured compositional and logical reasoning.
- Designing a neuro-symbolic framework for evaluating structured reasoning combining informal logic, logical structure, entailment modeling, and social consensus reasoning.
- Creating a commonsense QA benchmark with logically composed options and multi-step reasoning trees.
- Benchmarked reasoning quality using LLM prompting strategies and step-level inference quality.
- Targeting submission to **ACL or EMNLP 2026**.

**Responsible AI Researcher**, Institute of Cognitive Science, CU Boulder Jan 2025 – Present  
*Advisor:* [Dr. Theodora Chaspari](#), HUBBS Lab

- First-author (**IEEE-EMBS 2025**; [arXiv:2509.25795](#)) on algorithmic bias in LLM-based depression detection.
- Led large-scale fairness evaluation of LLaMA, GPT-4, and other LLMs for mental health classification.
- Implemented in-context prompting strategies and fairness-aware losses to mitigate bias in low-resource settings.
- Developed diagnostic probes and metrics to quantify bias in model outputs, influencing model selection.

**Adapter-Based Multi-Agent Systems for Scalable LLM Inference** Aug 2025 – Present  
*Course:* *Systems for Machine Learning, University of Colorado Boulder*

- Building a modular serving system where each agent is represented by a dynamic, independently trainable LoRA adapter on top of a frozen base model.
- Designing infrastructure for dynamic adapter loading, chaining, caching, and orchestration.
- Measuring system metrics (latency, throughput) and quality trade-offs across adapter-based inference pipelines.
- Project focuses on scaling lightweight agents, serving efficiency, and modularity for real-world deployment.

## PUBLICATIONS

- Junias, Obed\***, et al. *Assessing Algorithmic Bias in Language-Based Depression Detection: A Comparison of DNN and LLM Approaches*. IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI), 2025. (Presented). [arXiv:2509.25795](#) (\*Presenting First Co-Author)
- Junias, Obed**, et al. “*Commonsense Reasoning via Logical Entailment Trees: Towards Neuro-symbolic Inference Alignment*.” (In Preparation) Target: *ACL / EMNLP 2026*.

## AWARDS AND RECOGNITION

- NSF-EMBS-Google Young Professional NextGen Scholar**, *IEEE EMBS BHI Conference*, 2025

## SKILLS

|                    |                                                    |
|--------------------|----------------------------------------------------|
| Languages          | Python, C++, SQL                                   |
| AI/ML Frameworks   | PyTorch, TensorFlow, Transformers, LangChain, CUDA |
| Tools & Platforms  | Git, Docker, Kubernetes, Redis, MongoDB, Selenium  |
| Cloud Technologies | GCP, Oracle Cloud (OCI), AWS                       |

## RELEVANT PROJECTS

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- **Medical Ethics Assessment of LLMs**

- **Objective:** Assessed the ethical reasoning capabilities of large language models (LLMs) in clinical contexts, an underexplored dimension of Responsible AI.
- **Approach:** Designed a controlled multiple-choice evaluation pipeline using Retrieval-Augmented Generation (RAG) to measure the effect of external knowledge on ethical reasoning.
- **Findings:** Through logit-based confidence and rationale analysis, revealed that while LLMs show partial moral understanding, they lack reliability and depth for ethically sensitive tasks.

- **Lost in Plot: Contrastive Learning for Movie Retrieval**

- **Objective:** Developed a system for “tip-of-the-tongue” movie retrieval, addressing vague natural language queries.
- **Approach:** Fine-tuned a BERT encoder with a contrastive multi-task loss aligning user descriptions with movie metadata in a shared semantic space.
- **Result:** Outperformed a GPT-4 few-shot baseline on Recall@K and MRR, creating a new synthetic dataset of 100K+ movie descriptions.

- **Resource-Efficient LLM Fine-tuning for Mental Health Support**

- **Objective:** Adapted a large language model for mental health conversation support under limited computational and data resources.
- **Approach:** Applied parameter-efficient fine-tuning (PEFT) with Quantized Low-Rank Adaptation (QLoRA) to the Falcon 7B model, improving task specialization efficiency.

- **GradCompass: AI-Powered Graduate Application Assistant**

- **Objective:** Built a multi-agent LLM system to deliver personalized graduate school guidance, document evaluation, and recommendations.
- **Approach:** Implemented an **8-agent framework** with role-specific LLMs for university recommendation and advising, integrated with a RAG pipeline and an AI-based interview simulator.

- **E2E Text-to-Comic Generation Pipeline**

- **Objective:** Created an end-to-end web system that transforms textual narratives into comic strips.
- **Approach:** Integrated GPT-3.5 and Stable Diffusion via a modular pipeline orchestrated on Google Kubernetes Engine (GKE) with Redis caching for high-throughput, concurrent generation.

## INDUSTRY EXPERIENCE

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**Senior Member of Technical Staff** at Oracle Corporation

Aug 2021 - Aug 2024

- Architected and implemented **TestNG** and **BATS** automation frameworks, focusing on scalable and robust end-to-end testing for [Oracle Rest Data Services](#) and [SQLcL](#), reducing manual effort by 95%.
- Engineered a Python-based migration framework to automate the transfer of [Oracle APEX](#) applications, improving deployment efficiency and supporting **over 200 employees**.
- **Led a 15-member team** to validate a new [DB Tools](#) feature on Oracle Cloud in 2 months.

**Machine Learning Engineer Intern** at Hewlett Packard Enterprise

Feb 2021 - Jul 2021

- Built an intelligent automation solution with Groovy and [Workfusion](#) OCR, optimizing data workflows and integrating text extraction into software robots, **increasing speed by 40% and reducing errors by 80%**.
- Delivered a Proof of Concept (PoC) on Intelligent Business Process Management Software tools, showcasing automation potential to stakeholders.

## RESEARCH INTERESTS

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Natural Language Processing • Commonsense Reasoning • Neuro-symbolic NLP • Responsible Machine Learning • Logical Inference • Fairness and Interpretability in LLMs • LLM Systems and Scalability • Multi-Agent AI Architectures